

## A012 · Uncovering Heteroxylan Biosynthesis in Rice through Network-Based Gene Discovery and Protein Interaction Analysis

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**Presenter:** Mohsin Ali Nasir — Ohio University

**Authors:** Mohsin Ali Nasir, Samia Nawaz, Ahmed Faik

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Plant cell walls (PCWs) give mechanical strength, flexibility and protection to plant tissues. Heteroxylan (HX) is a major wall polysaccharide in grasses including rice that is necessary for tissue strength and development, but the genes and protein complexes regulating its synthesis remain unknown. Here, we report the identification and characterization of HX-related gene complexes in rice by an integrated genomics and functional validation approach. We created PlantNetX, a rice-centric gene co-expression platform, leveraging bulk RNA-seq data from 70 rice studies to support our effort. PlantNetX combined with gene association network (GAN) analysis revealed putative HX biosynthetic complexes including GT43 and GT47 glycosyltransferase genes, which have been shown to contribute to xylan biosynthesis. To support these predictions, we integrated 9 single-cell RNA-seq datasets into PlantNetX to assess candidate genes in various rice cell types and to investigate if predicted interacting genes are co-expressed in cell types involved in cell wall production. The NAPPA/i-GT-ray system is a high-throughput platform to screen for protein-protein interactions and is used to test for predicted protein interactions. Two predicted GT43/GT47 complexes were initially tested and showed positive interaction signals, proving the effectiveness of this technique for HX-related protein complex validation. The CRISPR/Cas9 gene editing system was also used to investigate specific GT43 genes in Kitaake rice. Mutant lines for GT43-D, GT43-H and GT43-E have been developed and are in characterization and their characteristics demonstrate altered growth, stem morphology, seed development and probable cell wall alterations comparable to previously reported xylan biosynthesis mutants. The PlantNetX-based co-expression analysis, GAN analysis, single-cell transcriptomics, protein interaction tests and CRISPR mutant characterization provide a framework for the identification of HX biosynthesis complexes and better understanding of plant cell wall construction.

## **Title: Uncovering Heteroxylan Biosynthesis in Rice through Network-Based Gene Discovery and Protein Interaction Analysis**

Mohsin Ali Nasir<sup>1,2</sup>, Samia Nawaz<sup>1</sup> and Ahmed Faik<sup>1,2</sup>

<sup>1</sup>Environmental and Plant Biology Department, Ohio University, Athens, Ohio, USA

<sup>2</sup>Interdisciplinary Program in Molecular and Cellular Biology, Ohio University, Athens Ohio, USA

### **Abstract**

Plant cell walls (PCWs) give mechanical strength, flexibility and protection to plant tissues. Heteroxylan (HX) is a major wall polysaccharide in grasses including rice that is necessary for tissue strength and development, but the genes and protein complexes regulating its synthesis remain unknown. Here, we report the identification and characterization of HX-related gene complexes in rice by an integrated genomics and functional validation approach. We created PlantNetX, a rice-centric gene co-expression platform, leveraging bulk RNA-seq data from 70 rice studies to support our effort. PlantNetX combined with gene association network (GAN) analysis revealed putative HX biosynthetic complexes including GT43 and GT47 glycosyltransferase genes, which have been shown to contribute to xylan biosynthesis. To support these predictions, we integrated 9 single-cell RNA-seq datasets into PlantNetX to assess candidate genes in various rice cell types and to investigate if predicted interacting genes are co-expressed in cell types involved in cell wall production. The NAPPA/i-GT-ray system is a high-throughput platform to screen for protein-protein interactions and is used to test for predicted protein interactions. Two predicted GT43/GT47 complexes were initially tested and showed positive interaction signals, proving the effectiveness of this technique for HX-related protein complex validation. The CRISPR/Cas9 gene editing system was also used to investigate specific GT43 genes in Kitaake rice. Mutant lines for *GT43-D*, *GT43-H* and *GT43-E* have been developed and are in characterization and their characteristics demonstrate altered growth, stem morphology, seed development and probable cell wall alterations comparable to previously reported xylan biosynthesis mutants. The PlantNetX-based co-expression analysis, GAN analysis, single-cell transcriptomics, protein interaction tests and CRISPR mutant characterization provide a framework for the identification of HX biosynthesis complexes and better understanding of plant cell wall construction.

**Keywords:** *Heteroxylan; rice; plant cell wall; PlantNetX; single-cell RNA-seq; GT43/GT47; CRISPR/Cas9; protein-protein interaction*